

UEC823: SOFT COMPUTING

L	T	P	Cr
2	0	2	3.0

Course objective: To familiarize with soft computing concepts. Introduce the ideas of neural networks, fuzzy logic and use of heuristics based on human experience. Familiarize the concepts of Genetic algorithm. Apply the soft computing concepts to solve practical problems.

Introduction: Introduction to soft computing, Problem complexity, Problem complexity classification, Types of soft computing techniques, Softcomputing versus hard computing, Advantages of soft computing.

Artificial Neural Networks: Biological neuron, Artificial Neural Network, Mathematical Models, McCulloch Neural Model, Perceptron, Adaline and Madaline, Learning & Training in ANN, Hopfield Neural Network, Self-Organizing Networks, Recurrent Networks, Associative memories

Fuzzy Logic System: Crisp Vs Fuzzy set theory, Membership functions, Fuzzy set operations, Fuzzy rules, Mamdani and Sugeno fuzzy inference systems, Defuzzification methods.

Genetic Algorithms: Introduction and biological background of GA, String Encoding of chromosomes, Selection methods, Single & multi-point crossover operation, Mutation, Adjustment of strategy parameters such as Population size, Mutation & Crossover probabilities

Familiarization with standards: IEEE 2894-2024: IEEE 1872.1-2024, IEEE 7000-2021, IEEE 1855-2016, IEEE 1918.1.1-2024.

Laboratory Work: MATLAB Toolboxes: Fuzzy Logic Toolbox, Neural Network Toolbox, Neural network as a classifier, FLS for Antilock Braking System (ABS), GA in route planning for Travelling Sales Person, Time-Series forecasting using ANN. Familiarization of GA toolbox MATLAB and implementing it to find optimal solution of optimization problems.

Course Learning Outcomes: Upon completion of this course, the student should be able to:

1. Identify the characteristics of Soft Computing Techniques
2. Analyze neural networks and their applications.
3. Apply fuzzy logic and fuzzy reasoning for decision making
4. Analyze genetic algorithms and their applications.

Text Books

1. Jang, J.S.R., Sun, C.T., and Mizutani, E., Neuro-Fuzzy and Soft Computing, Pearson Education (2004) 2nd ed.
2. Eberhart, R., Simpson, P., and Dobbins, R., Computational Intelligence - PC Tools, AP Professional (1996) 3rd ed.

Reference Books:

1. Jacek M. Zurada – Introduction to Artificial Neural Systems
2. S N Sivanandam, S N Deepa – Principles of Soft Computing, Wiley Publications
3. John Yen, Reza Langari – Fuzzy Logic Intelligence, Control, and Information
4. Goldberg, Davis E., Genetic Algorithms: Search, Optimization and Machine Learning, Wesley Addison (1989) 3rd ed